

PHANTOM ALIGNMENT PROCEDURE

Use the following procedure to physically center the phantom within the phantom box. Once this is accomplished, centering the phantom in the field should only require the Service Engineer to place the table top at the positions recorded in this procedure.

1. Place the phantom box on the table top so that the front of the phantom box is flush with the end of the table top. Place the phantom in the phantom holder.
2. Align the phantom in the field using the laser so that the 8" water phantom is in the beam.
3. Scan the phantom using the center cal or H-F (half-field) water protocol from the Test Protocols screen of the User Interface. Use the default mA, no comp.
4. *[Need phantom check procedure here.](#)*

LEFT/RIGHT ADJUSTMENT

5. Loosen the screws on **one** of the phantom holder wedges so that the phantom can be moved in the required direction. Refer to Figure Figure 76.
6. Move the wedge the desired distance, tighten the screw and then slide the phantom into its new location.

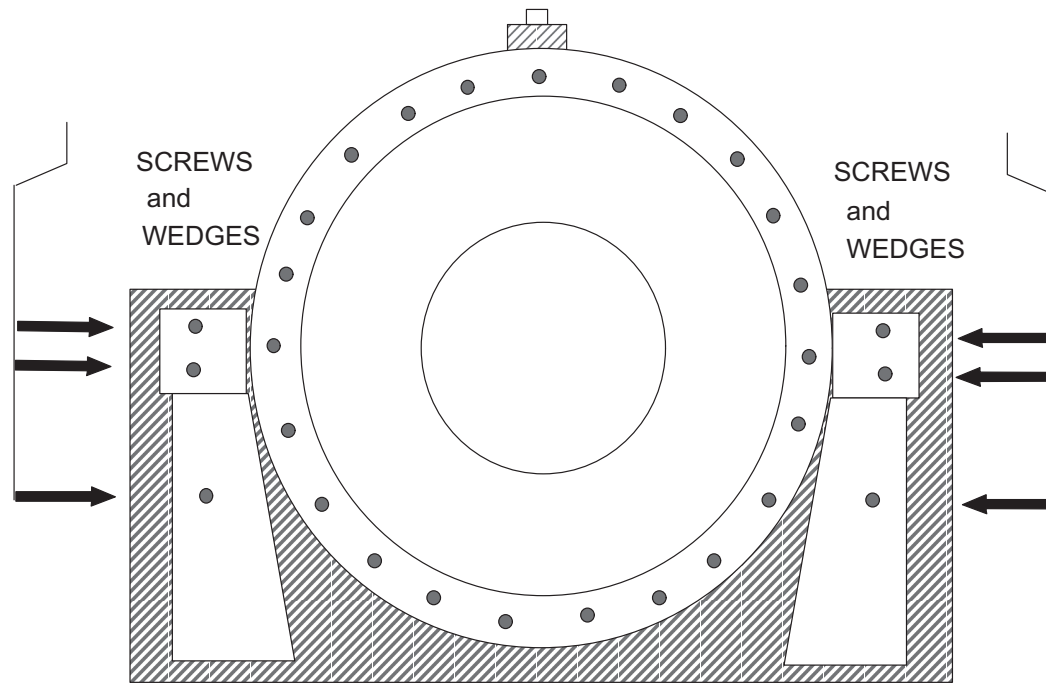


FIGURE 76 MOUNTING SCREWS FOR PHANTOM WEDGES

UP/DOWN ADJUSTMENT

7. Use the Gantry control panels to move the table top into the correct vertical location.
8. Re-scan the phantom as many times as needed to adjust the phantom to proper centering position. When the phantom is within the ± 1 mm specification of the **check phantom(??)** program, record the table top horizontal and vertical locations and use these locations when it is necessary to scan a centered phantom.

NEW PHANTOM TEST KIT *(STARTING SEPTEMBER 1997)*

A Beam Analysis kit has been provided with the scanner systems. This kit is used to verify the calibration and performance of the system. The kit is also used as part of regularly scheduled quality control checks.

The test kit contains: phantom, phantom support, centering pin and two bottles of phantom refill solution. The illustration shows the test kit as it appears packed in its shipping and storage box. The phantom support is always the topmost piece in the kit.

The **phantom support** attaches to the patient support and positions the phantom and centering pin.

The **phantom** is used for evaluating image quality and has five separate scannable sections:

12" water... used for full field.

8" water... used for half field.

6" water... used for low contrast measurements.

6" low contrast... used for low contrast measurements.

6" slice thickness/MTF.

Proper scanner calibration requires the phantom have no air bubbles in any scan section. Additional distilled water can be added to the phantom via the expansion chamber which should be 3/4 filled.

The **centering pin** is used for calibrating global and individual centering constants.

NEW PHANTOM MOUNTING PROCEDURE

1. Remove the phantom support from the storage box.
2. Loosen the adjustable foot knob to its maximum open position by turning the knob counterclockwise. Refer to Figure 77.

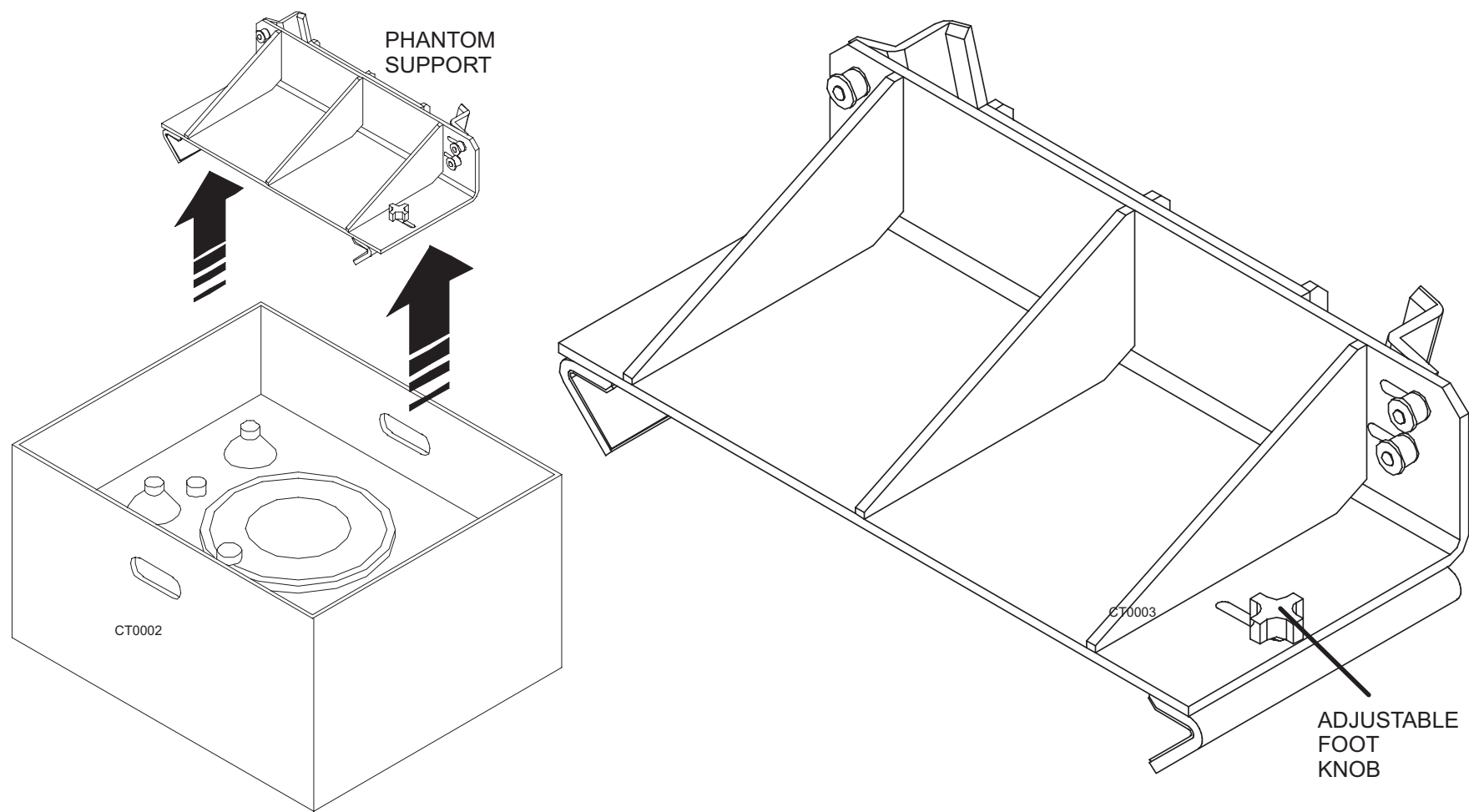


FIGURE 77 NEW PHANTOM SUPPORT

3. Stand on the right side of the patient couch. Place the phantom support on the couch by sliding the supports' feet onto the couch until the phantom support overhang on the front of the support butts up flush against the front of the patient couch. Refer to Figure 78.

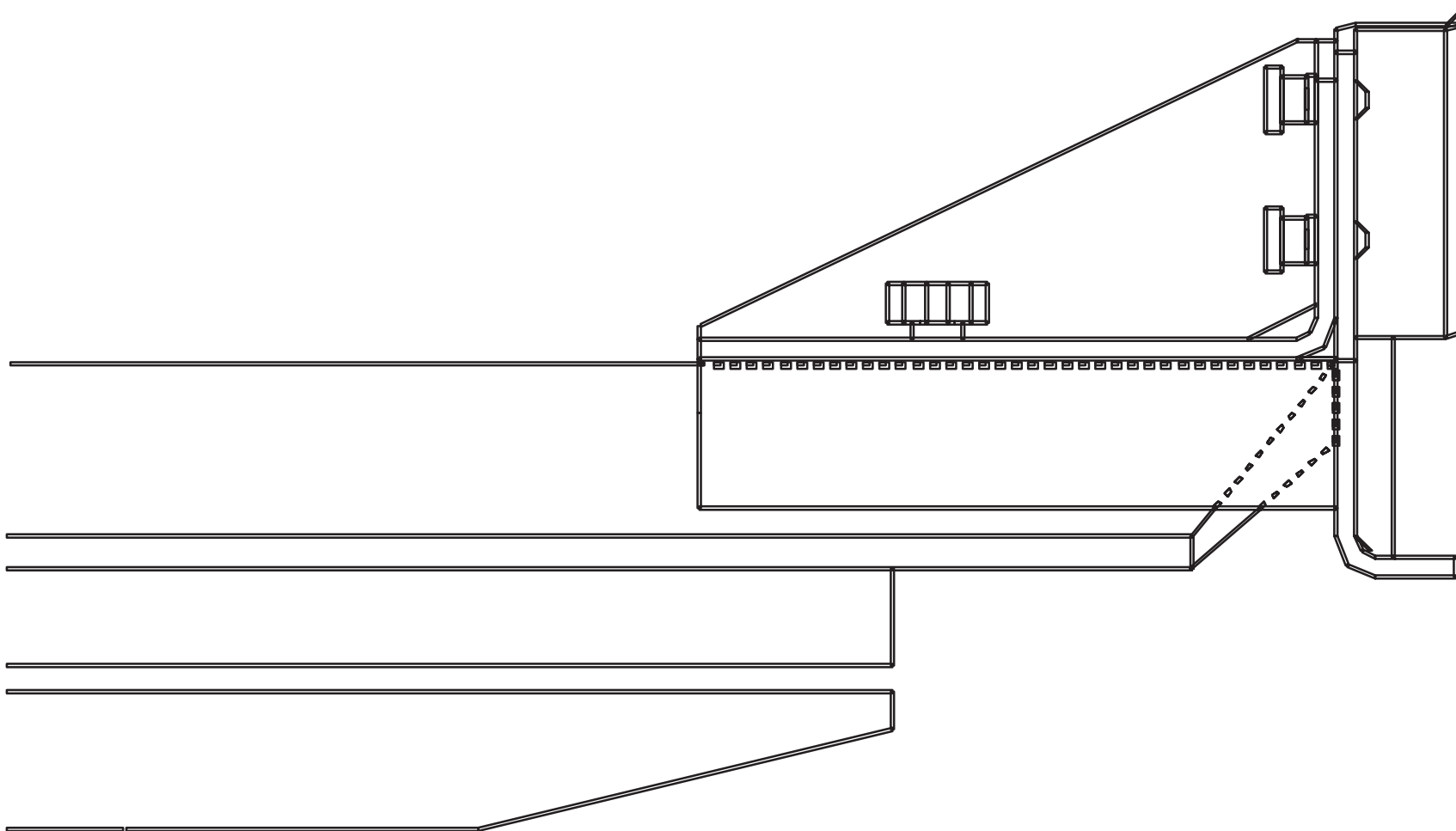


FIGURE 78 NEW FLUSH MOUNTED PHANTOM SUPPORT

4. With both feet underneath the edge of the couch top and the front of the support flush to the patient couch, place your left hand on the left corner of the support and slide unit toward you. Make sure the front overlay piece of the support remains flush to the front of the couch. Refer to Figure 79.
5. The front and left side of the phantom support should now fit snug to the patient couch. Hold the phantom support in place with your left hand. With the right hand, push the right foot bracket in toward the patient couch and tighten the right foot by turning the adjustable knob clockwise until the foot fits snug to the right side of the patient support. Refer to Figure 79.

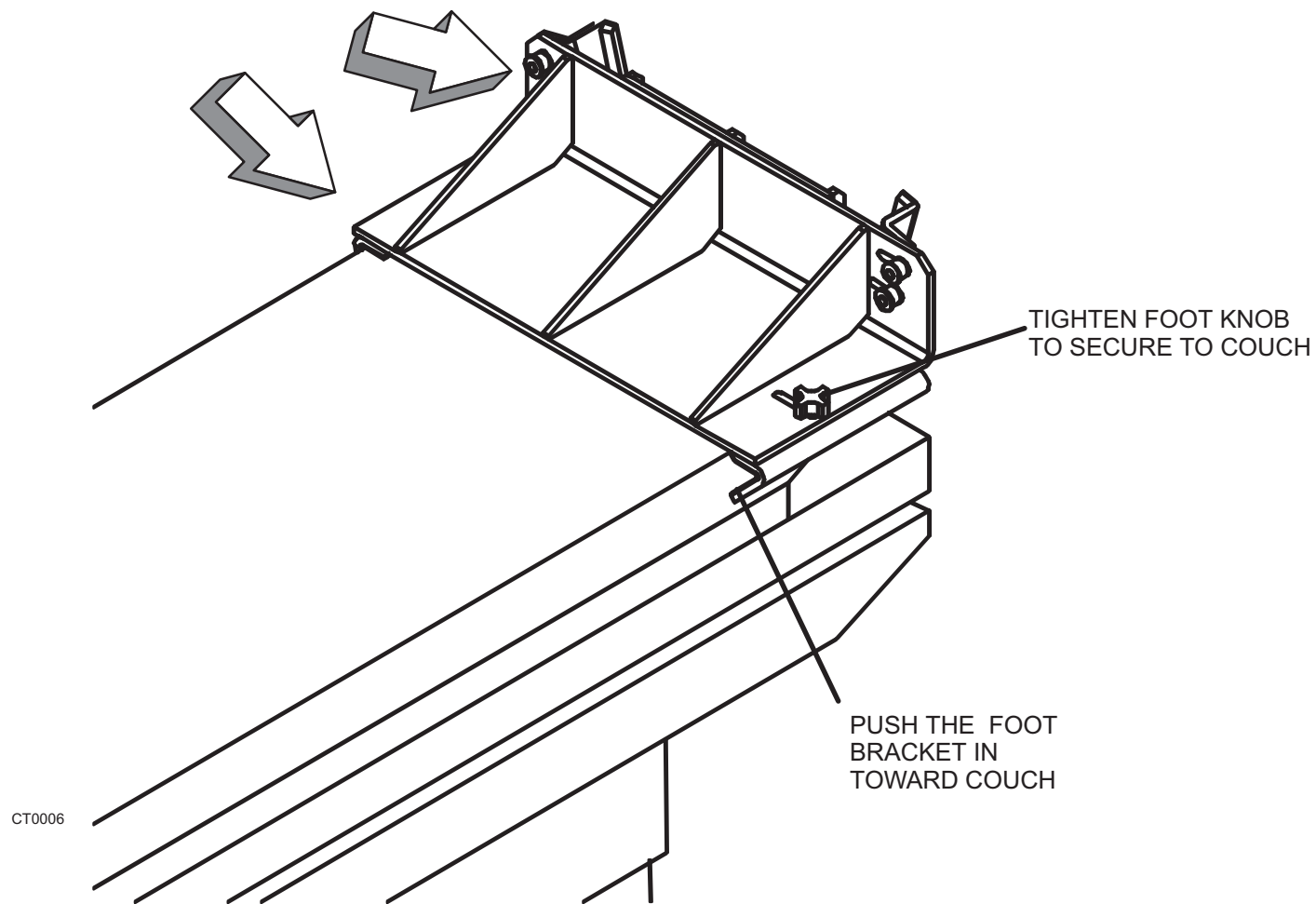


FIGURE 79 SECURING NEW PHANTOM SUPPORT TO PATIENT SUPPORT

Place New Phantom in Retainer Brackets

6. Remove the phantom from the storage box and mount it on the phantom support by inserting the phantom's flange (the part with the aluminum screw posts) behind the retainer holding brackets and position the overflow chamber at the top. Make sure the phantom rests securely between the left/right adjustable holding brackets by turning the screws to tighten brackets.

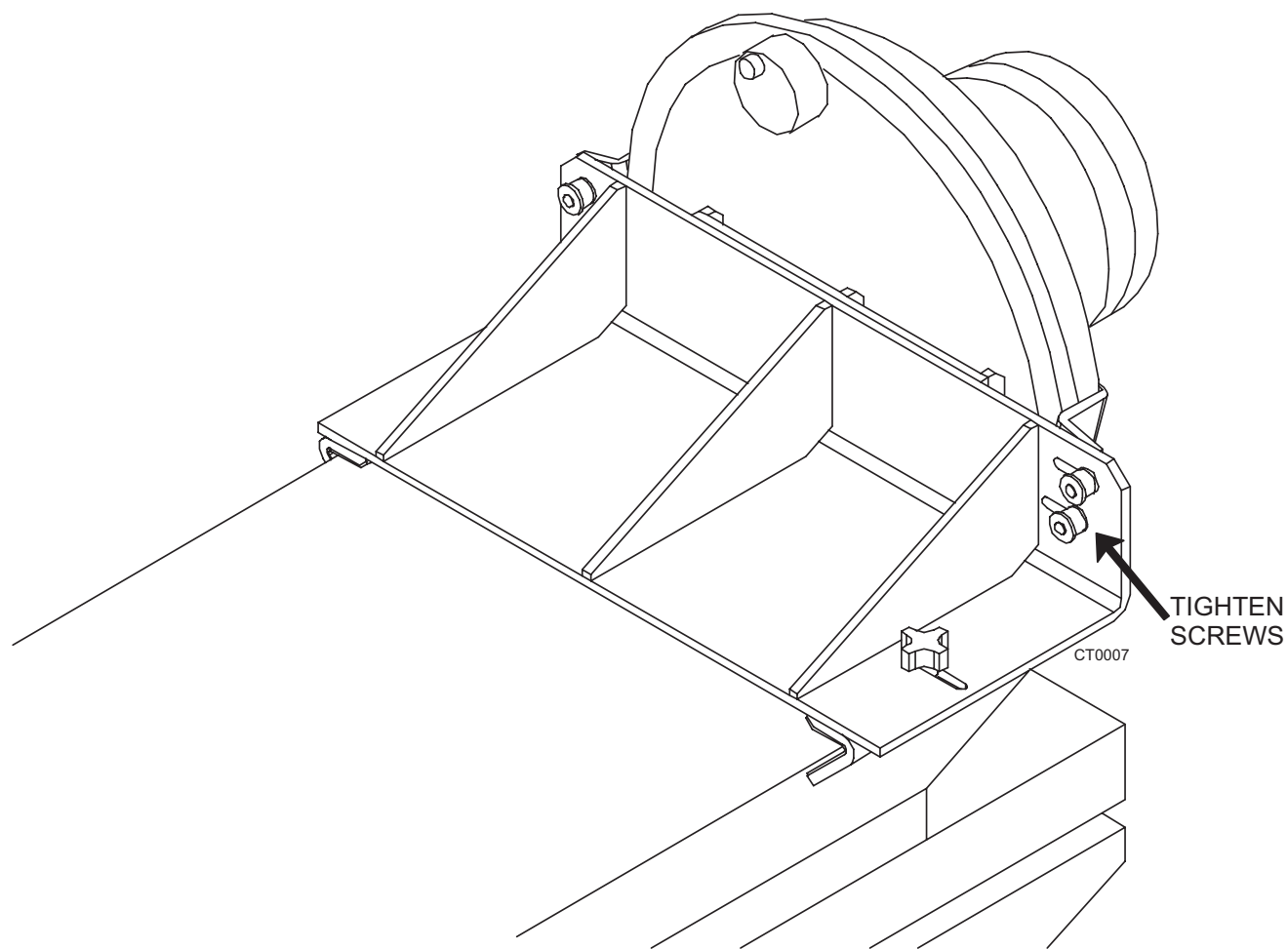


FIGURE 80 PHANTOM MOUNTED ON PATIENT SUPPORT

7. With the phantom sitting in the support, adjust the left/right centering by loosening the screws and sliding the brackets as needed in their slots. Turn the screws clockwise to re-tighten them after making the centering adjustment and rescan to see if centered in the field of view. Refer to Figure 81.

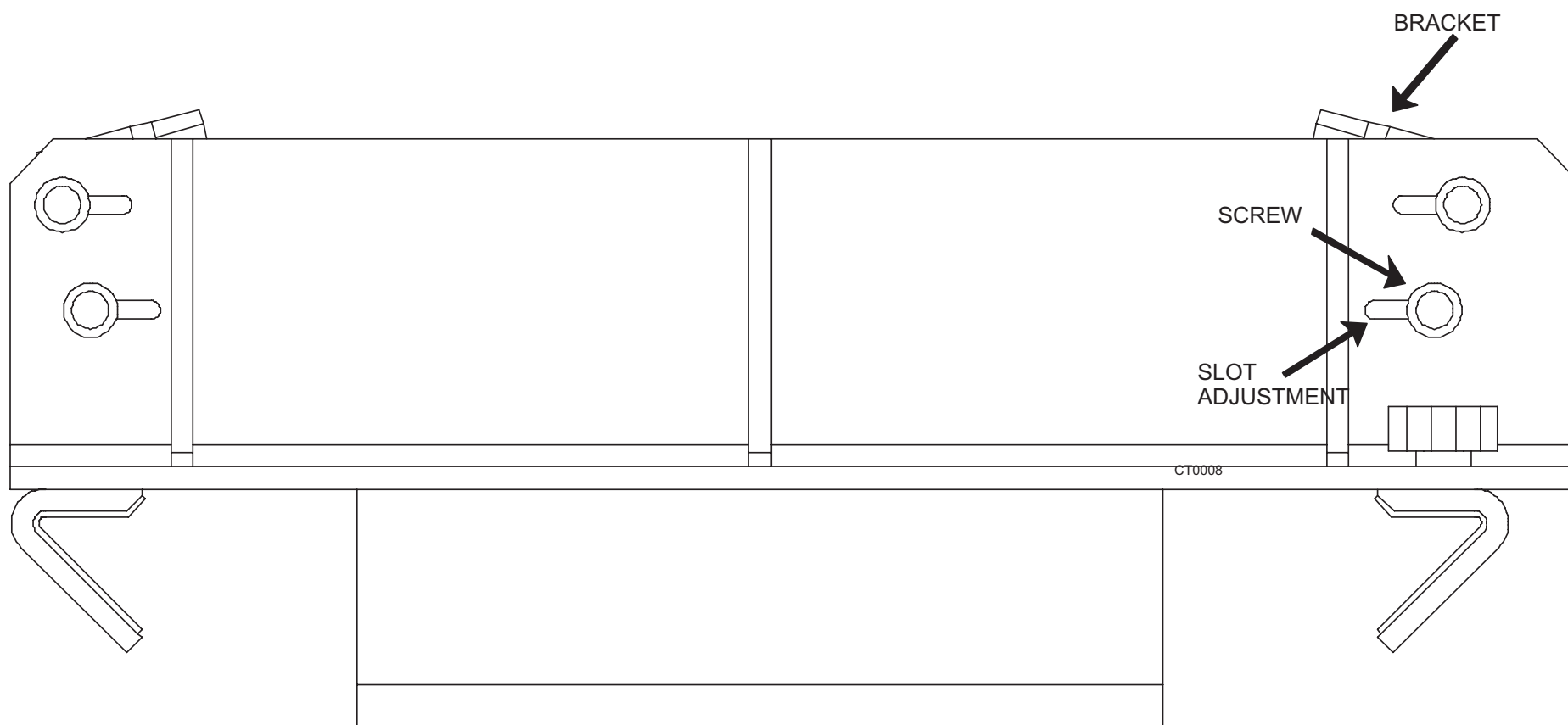


FIGURE 81 RETAINER BRACKET ADJUSTMENT SLOTS

The phantom should now be securely held between the retainer brackets on the phantom support.

Phantom Support and Phantom Positioning

The phantom is properly positioned for the scan when the phantom support is flush with the end of the patient couch, fits snugly to the sides of the patient couch, the phantom overflow chamber is at the top and the centering pin is inserted. Refer to Figure 82.

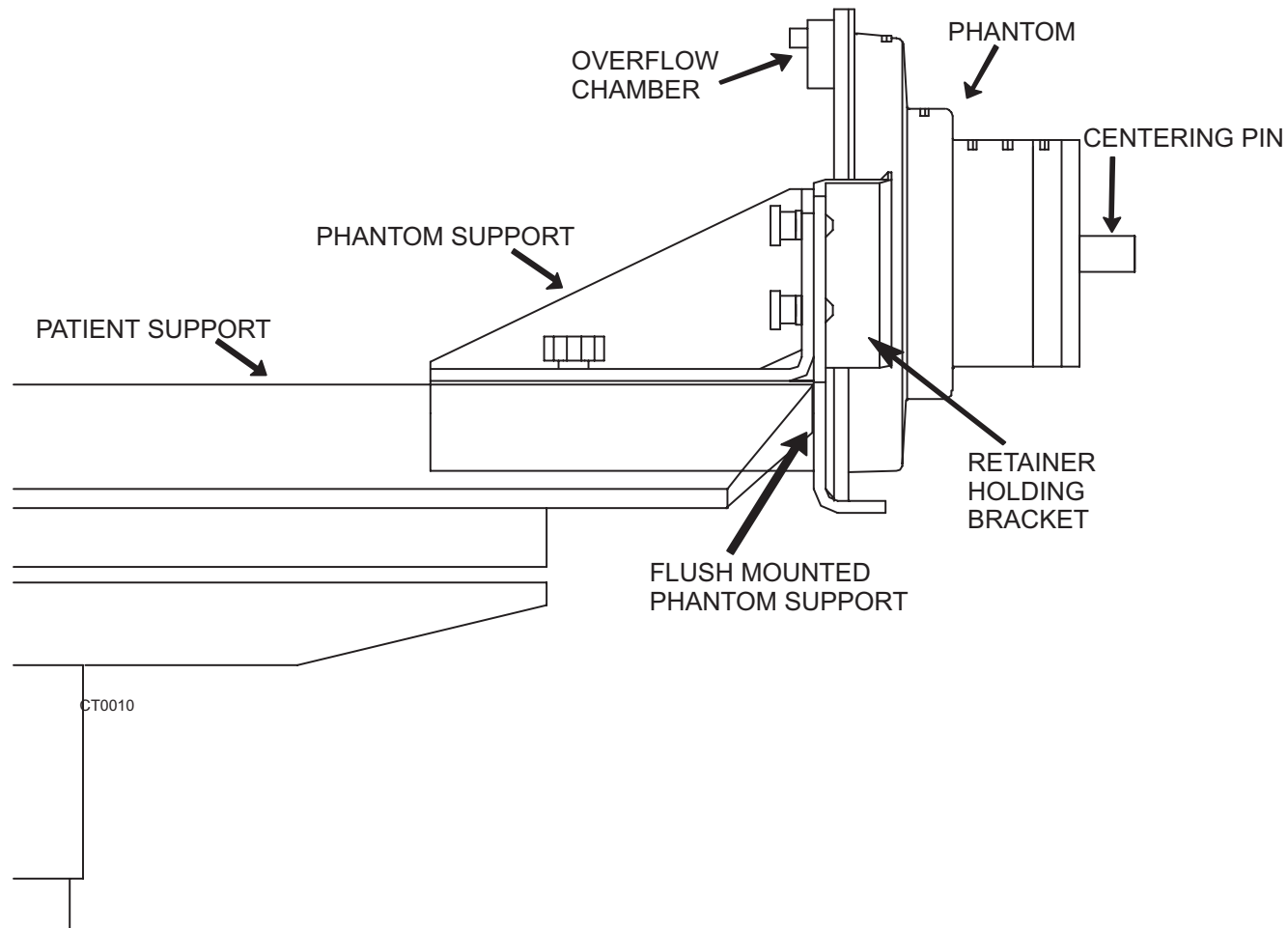


FIGURE 82 PROPERLY MOUNTED SUPPORT AND PHANTOM